Bangladesh University of Business and Technology

LAB REPORT

Experiment Date : 24/10/2025
Experiment No : 02
Course Title : Structured Programming Language Lab
Course Code : CSE-102
Experiment Name: Lab report (Array)

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1) Write a C program to read and print elements of array – using recursion.

Code:

```c
#include <stdio.h>

void readArray(int a[], int i, int n) {
    if (i < n) {

        printf("Enter element %d: ", i + 1);
        scanf("%d", &a[i]);
        readArray(a, i + 1, n);
    }
}

void printArray(int a[], int i, int n) {
    if (i < n) {
        printf("%d ", a[i]);
        printArray(a, i + 1, n);
    }
}

int main() {

    int a[50], n;

    printf("Enter size of array: ");

    scanf("%d", &n);

    readArray(a, 0, n);

    printf("Array elements are: ");

    printArray(a, 0, n);

    return 0;
}
```

Output:

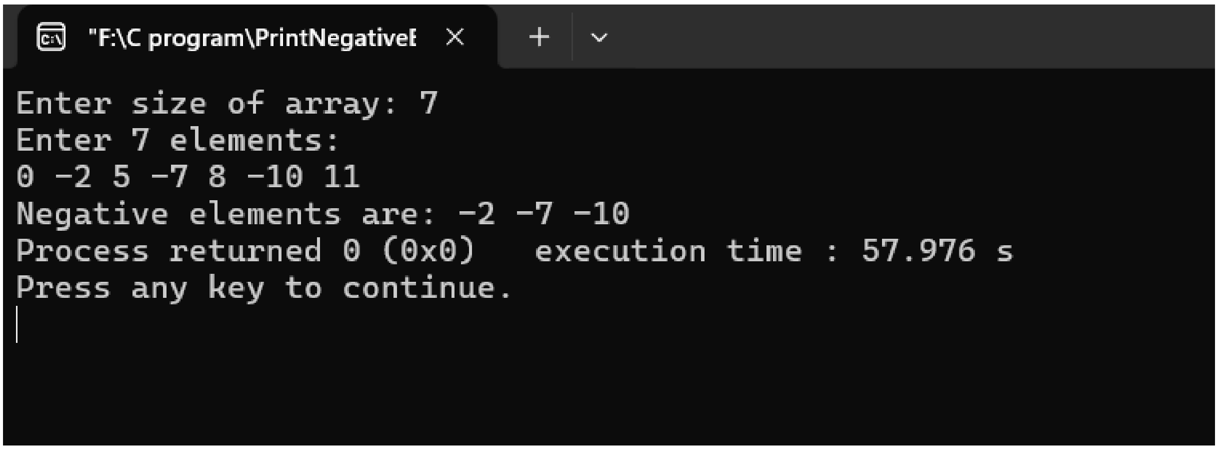
```
"F:\C program\Print ElementC

Enter size of array: 5
Enter element 1: 50
Enter element 2: 60
Enter element 3: 70
Enter element 4: 80
Enter element 5: 90
Array elements are: 50 60 70 80 90
Process returned 0 (0x0)   execution time : 9.920 s
Press any key to continue.
```

2. Write a C program to print all negative elements in an array.

Code:

```
#include <stdio.h>

int main() {

    int a[50], n, i;


    printf("Enter size of array: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
     for (i = 0; i < n; i++)


       scanf("%d", &a[i]);
    printf("Negative elements are: ");



    for (i = 0; i < n; i++) {
        if (a[i] < 0)
            printf("%d ", a[i]);

    }
    return 0;
}
```

Output:



3. Write a C program to find sum of all array elements – using recursion.

Code:

```c
#include <stdio.h>


int sumArray(int a[], int i, int n) {
    if (i < n)
        return a[i] + sumArray(a, i + 1, n);
    else
        return 0;
}


int main() {
    int a[50], n, i;


    printf("Enter size of array: ");
    scanf("%d", &n);


    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);


    int sum = sumArray(a, 0, n);
    printf("Sum of all elements = %d", sum);

    return 0;
}
```

Output:

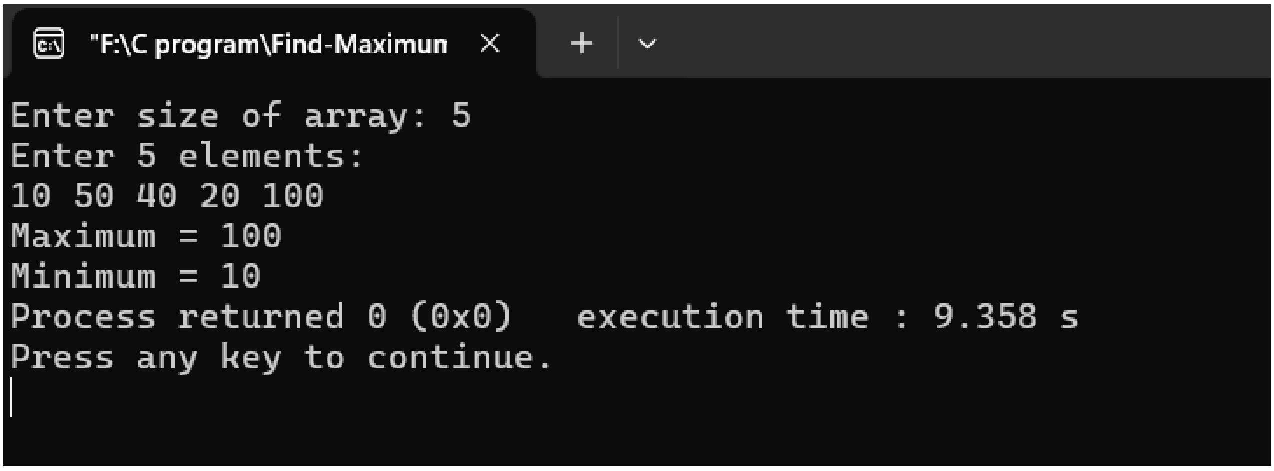
```
"F:\C program\SumOfAllElem

Enter size of array: 10
Enter 10 elements:
10 20 30 40 50 60 70 80 90 100
Sum of all elements = 550
Process returned 0 (0x0)   execution time : 12.094 s
Press any key to continue.
```

4) Write a C program to find maximum and minimum element in an array – using recursion.

Code:

```
#include <stdio.h>

int findMax(int a[], int i, int n) {
    if (i == n - 1)
        return a[i];
    int max = findMax(a, i + 1, n);
    return (a[i] > max) ? a[i] : max;
}

int findMin(int a[], int i, int n) {
    if (i == n - 1)
        return a[i];
    int min = findMin(a, i + 1, n);
    return (a[i] < min) ? a[i] : min;
}

int main() {
4

    int a[50], n, i;
    printf("Enter size of array: ");
    scanf("%d", &n);

    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Maximum = %d\n", findMax(a, 0, n));
    printf("Minimum = %d", findMin(a, 0, n));

    return 0;
}
```

Output:

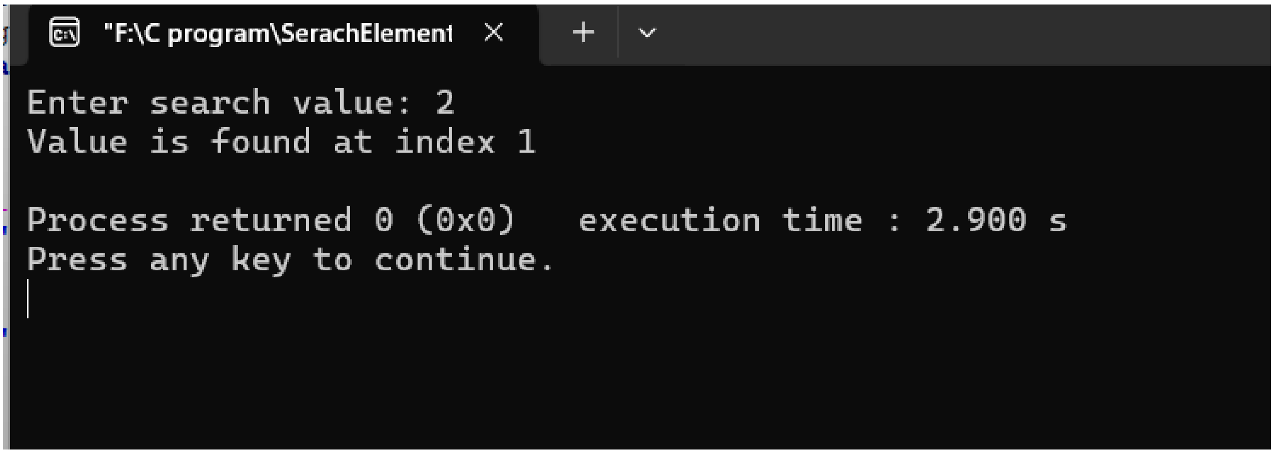


5) Write a C program to search an element in an array using linear search.

Code:

```c
#include <stdio.h>

int main() {

    int a[10] = {10, 2, -1, 0, 3, 7, -5, 9, 4, 6};
    int search_value, i, flag = 0;

    printf("Enter search value: ");


    scanf("%d", &search_value);

    for(i = 0; i < 10; i++) {
        if(search_value == a[i]) {
            flag = 1; // found
            break;
        }
    }


    if(flag == 1)
        printf("Value is found at index %d\n", i);


  else


        printf("Value not found\n");

    return 0;
}
```

Output:



"F:\C program\SerachElement

```
Enter search value: 7
Value is found at index 5

Process returned 0 (0x0)   execution time : 1.610 s
Press any key to continue.
```

"F:\C program\SerachElement

```
Enter search value: 6
Value is found at index 9

Process returned 0 (0x0)   execution time : 1.365 s
Press any key to continue.
```